HAMZA ALMUGHRABI

ML ENGINEER & DATA SCIENTIST & DATA ANALYST

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SUMMARY

AI & Robotics Engineering diploma graduate with hands-on project experience in machine learning, data science, and analytics. Skilled in Python, SQL, and Power BI, with projects covering data preprocessing, predictive modeling, RAG assistants, and interactive dashboards. Interested in applying AI and data techniques to build practical solutions that support business decisions.

PROFESSIONAL SKILLS

Data Science & Architecture:

- Machine Learning (Regression, Classification, Clustering)
- Deep Learning (Neural Networks, CNN, RNN, NLP Basics)
- AI Agents & Workflow Automation (n8n Basics)
- Prompt Engineering & AI Solution Building (RAG, AI Workflows)
- Data Analytics & Modeling
- Data Architecture & Data Warehousing
- Data Cleaning, Transformation & ETL/ELT Pipelines
- Feature Engineering & Data Preprocessing
- Data Analytics, Reporting & Dashboards (Power BI, Matplotlib, Seaborn, Excel)

Programming & Databases:

- Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch basics)
- SQL (Query Optimization, MS Access)
- Cloud & Big Data: Databricks, GCP Basics (BigQuery)
- Web Technologies (HTML, CSS, JavaScript, React.js Basic, PHP Basic, Arduino, C++)

Tools & Productivity:

- GitHub, Jupyter Notebooks, Google Colab, VS Code
- Microsoft Excel, Microsoft Word, Microsoft PowerPoint, Figma, Canva

EDUCATION

Intermediate Diploma in AI & Robotics Engineering (2022-2025)

Al-Balqa Applied University

- Machine Learning, Computer Vision, Data Structures.

CERTIFICATION

- Machine Learning (The Hope International)
- Deep Learning (The Hope International)
- Command Engineering for ai (Dubai future foundation)
- Preparing Data for Analysis with Microsoft Excel (Microsoft)
- Harnessing the Power of Data with Power BI (Microsoft)
- Extract, Transform and Load Data in Power BI (Microsoft)
- Data Modeling in Power BI (Microsoft)
- Data Analysis and Visualization with Power BI (Microsoft)
- Creative Designing in Power BI (Microsoft)
- Foundations: Data, Data, Everywhere (Google Cloud)
- Build Data Lakes and Data Warehouses (Google Cloud)
- Machine Learning with PyTorch (IBM Developer)
- Intermediate Python for Developers (DataCamp)
- Supervised Learning with scikit-learn (DataCamp)
- Web Design (Basic) (HTU)
- Amazon SageMaker AI Getting Started (AWS)

PROJECT EXPERIENCE

CAR INSURANCE CLAIM PREDICTION SYSTEM (2026 - Present)

Python, Scikit-Learn, Streamlit, Machine Learning | GitHub, Streamlit

- Developed an end-to-end machine learning system using an optimized **Soft Voting** Classifier (**Random Forest & Gradient Boosting**) with **GridSearchCV** to predict car insurance claims.
- Built and deployed an interactive **Streamlit web application** with real-time risk assessment, automated feature alignment, and modular project structure.

TELECOM CUSTOMER CHURN ANALYSIS SYSTEM (2025)

Advanced Excel, Pivot Tables, Data Visualization | GitHub

- Processed and analyzed a telecom dataset of **6,687 customers** to identify key drivers behind a **26.86% churn rate** using advanced data aggregation and feature engineering.
- Developed an interactive **Excel dashboard** to segment churn by demographics (44% senior churn) and geography, capturing a critical **63.24% churn spike** in California.

DATA WAREHOUSE & ANALYTICS SYSTEM (2026)

SQL Server, Data Modeling | GitHub

- Designed a **Data Warehouse** using **SQL Server** and **Databricks** (**Medallion Architecture**) to centralize ERP & CRM sales data.
- Executed robust ETL data cleaning and delivered SQL-based analytics to uncover critical insights into customer trends and product performance.

FINANCIAL RAG AI ASSISTANT SYSTEM (2026)

Python, LangChain, Streamlit, ChromaDB, Groq API, Llama 3.1-8b-instant | GitHub

- Deployed an end-to-end **RAG pipeline** to automate financial queries, optimizing data ingestion via parsing, dynamic chunking of PDFs, and **ChromaDB** vector storage.
- Integrated **Llama-3.1-8b-instant** via Groq API with temperature set to 0.1, building an interactive **Streamlit UI** with source-verification tracking.

SALES FORECASTING SYSTEM (2025)

Python, Power BI, XGBoost | GitHub

- Built an **XGBoost** regression model (project evaluation: R^2 = 0.999, RMSE = 56.6) utilizing advanced feature engineering to forecast future sales trends.
- Developed an interactive **Power BI** dashboard to visualize sales drivers and deliver data-driven metrics for strategic business decisions.

WORKSHOP & JOB SIMULATION

- POWER BI_HR (Workshop)
- BCG X (Job Simulation)
- Your Real Guide to Entering (SHAI) (Workshop)

PROJECT EXPERIENCE (CONTINUED)

INSTAGRAM INTELLIGENCE (2026)
n8n, React, TypeScript, FastAPI, PostgreSQL
Built six n8n workflows for Instagram comment collection, AI classification, guarded replies and moderation, Telegram alerts and Google Sheets synchronization.
Implemented owner authentication, duplicate-action protection and an isolated public demo with fictional data.
Demo: https://instagram-intelligence-ui.onrender.com/?demo=1

LANGUAGES: Arabic (Native) | English (B1-B2)